

UNIT 04

Ethics, Privacy and Regulation

The rules, risks and responsibilities that come with using AI - and how to keep a human properly in charge of decisions.

Privacy

Security

Regulation

Decision Making

PRESENTED BY

Lingesh K Register No. **OSI2509030**

What we will cover

One rule ties the whole unit together: a person is always accountable for the outcome.

01 Why ethics is practical
It is a business risk, not a philosophy class

02 Privacy
What data you may collect, keep and use

03 Security
New attack surfaces that come with AI

04 Bias and fairness
How unfairness gets in, and how to catch it

05 AI regulations
The global picture in plain terms

06 Risk-based rules
Why the same tool is fine here and banned there

07 AI in decision making
Who decides, and who is answerable

08 A governance checklist
What good practice looks like day to day



01

Why This Matters Commercially

Ethics failures show up as fines, lawsuits and lost customers.

Four reasons this is not optional

Doing this well is a commercial advantage, not a cost centre.

Legal exposure

Fines under data and AI law are now a percentage of global revenue, not a small penalty.

Reputation

One unfair or leaked outcome can undo years of brand building.

Customer trust

People share data with organisations they believe will handle it properly.

Access to markets

Buyers and regulators increasingly ask how your AI works before they sign.

02

Privacy and Security

Protecting the data, and protecting the system.

Six principles that cover most of the law

These appear in GDPR, India's DPDP Act and most other privacy laws.

Purpose

Collect data for a stated reason and use it only for that.

Minimisation

Collect the least you need. Extra data is extra risk.

Consent

People should know and agree, in language they understand.

Accuracy

Wrong data produces wrong decisions about real people.

Retention

Delete it when the purpose ends. Do not keep it forever.

Rights

People can ask what you hold, correct it and have it removed.

How private data actually leaks with Gen AI

Most incidents are not attacks. They are ordinary people taking a shortcut.

The single most useful policy: never paste anything into a public AI tool that you would not post publicly.

- **Pasting into public tools**

Staff paste customer data or contracts into a free chatbot to save time.

- **Training on your input**

Some consumer tools may use what you type to improve the model.

- **Output that remembers**

A model can repeat sensitive detail it saw during training.

- **Screenshots and exports**

Data leaves your controlled systems and lands in someone's downloads folder.

- **Shadow AI**

Teams adopt tools nobody approved, so nobody can secure them.

- **Third-party add-ons**

A browser plugin can read everything on the page, including customer records.

New risks that come with AI systems

Old security still applies. These are the additions.

Prompt injection

Hidden instructions inside a document or web page hijack what the AI does next.

Data poisoning

Bad data is fed in deliberately so the model learns the wrong thing.

Model theft

Repeated querying is used to copy the behaviour of a model you paid to build.

Deepfakes

Cloned voice or video used to authorise a payment or impersonate a leader.

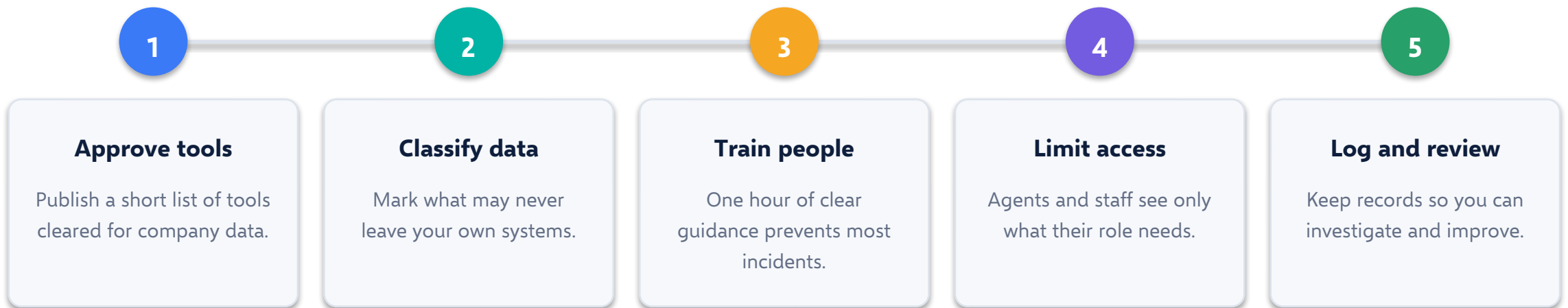
Over-permissioned agents

An agent connected to too many systems can do too much damage.

Supply chain

A vulnerability in a third-party AI service becomes your incident.

Five controls that prevent most problems



Banning AI does not work - people use it on their phones instead. Provide a safe approved option.

03

Bias and Fairness

The risk that is hardest to see and easiest to inherit.

Bias is learned, not programmed

Nobody writes an unfair rule. The model copies the pattern it was shown.

- Past decisions were made by people, with their biases
- The model learns those patterns as 'normal'
- Groups missing from the data get served worst
- The output looks neutral and statistical
- At scale, a small bias affects thousands of people

Representation bias

Some groups barely appear in the training data.

Historical bias

The past was unfair, so the prediction is too.

Proxy bias

A postcode or school name stands in for something protected.



Four practical fairness controls

You cannot remove bias completely. You can measure it and manage it.

Test by group

- Measure accuracy for each group separately
- An overall score can hide a serious gap
- Do this before launch and again later

Mixed review teams

- People who differ spot different problems
- Involve the affected function, not only IT
- Document what you checked

Explainability

- Be able to say which factors mattered
- If you cannot explain it, do not automate it
- Required by law in many high-risk uses

Appeal route

- A person can ask for a human review
- Someone has authority to overturn
- Track overturns - they show where it fails

04

AI Regulations

What the law now expects, in plain language.

The frameworks that matter most

Framework	Where	What it does	Why you care
EU AI Act	European Union	Sorts AI by risk level and bans the worst uses	The global benchmark - others copy it
GDPR	European Union	Governs personal data and automated decisions	Applies if you touch EU data
DPDP Act	India	Consent, purpose limits and duties for data handlers	Direct compliance duty in India
NIST AI RMF	United States	A voluntary framework for managing AI risk	Practical template for governance
Sector rules	Everywhere	Finance, health and hiring rules already apply	Existing law did not pause for AI

Details differ by country. The direction is the same everywhere: more transparency and more accountability.

Regulation is risk-based, not technology-based

The same technology can sit in any band. What matters is what you use it for.

L4

Unacceptable

Social scoring and manipulative systems. Banned outright.

L3

High risk

Hiring, credit, education, healthcare. Duty: testing, documentation, human oversight.

L2

Limited risk

Chatbots and generated content. Duty: tell people it is AI.

L1

Minimal risk

Spam filters, grammar help, product recommendations. Use freely.

Six duties that keep coming up

Disclose AI use

Tell people when they are interacting with AI or seeing generated content.

Keep documentation

Record what the system does, on what data, and how it was tested.

Provide human oversight

A person can review, override and stop the system.

Explain decisions

Be able to give a meaningful reason for an outcome that affects someone.

Protect the data

Lawful basis, security controls and defined retention.

Monitor after launch

Performance drifts. Review it on a schedule, not once.

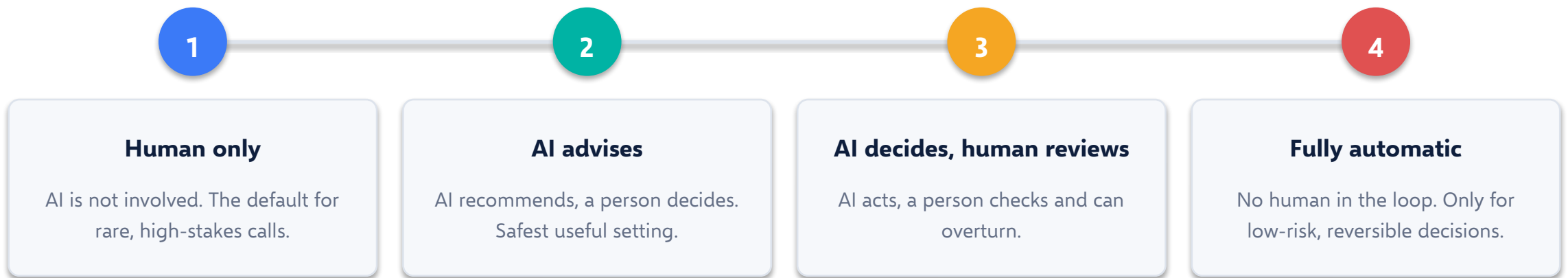
05

AI in Decision Making

The question that decides everything else: who is accountable?

Four ways AI can take part in a decision

Match the level to the consequence of being wrong.



Most business decisions belong in level 2. Most incidents happen when something quietly drifts to level 4.

Where automation is fine, and where it is not

SAFE TO AUTOMATE

Low stakes and easy to reverse

- Sorting and tagging incoming requests
- Suggesting products or content
- Drafting internal documents
- Flagging items for human review
- Routine, high-volume, low-value calls

KEEP A HUMAN

Life, money, rights or reputation

- Hiring, promotion and dismissal
- Credit, insurance and pricing decisions
- Medical and safety judgements
- Anything that is hard to undo
- Anything a person could reasonably appeal

IN SHORT Ask one question: if this is wrong, who is harmed and can we undo it?

What 'a human in the loop' has to mean

A rubber stamp is not oversight. It is just paperwork.

- **Real authority**

The reviewer can actually say no, without needing permission.

- **Enough time**

Sixty decisions an hour is not review, it is rubber-stamping.

- **Enough information**

The reviewer sees why the AI suggested this.

- **Named accountability**

One person owns the outcome. Not 'the algorithm'.

- **Overtures are tracked**

Frequent overrides are a signal the model needs work.

- **Training on limits**

Reviewers know where the AI is usually wrong.

THE PRINCIPLE TO REMEMBER

**A model can make a recommendation. Only
a person can be accountable.**

This is the line that regulators, courts and customers all care about.

A practical starting checklist

Governance is boring, cheap and the reason projects survive.

Write a short AI policy

Two pages people will actually read:
approved tools, banned data, who to ask.

Keep an AI register

A list of every AI use, its owner and its risk level.

Classify each use case

Minimal, limited or high risk - and apply the matching controls.

Review on a schedule

Quarterly checks on accuracy, fairness and complaints.

Train everyone once a year

Short, practical, example-led. Not a legal lecture.

Have an incident route

People know exactly who to tell when something goes wrong.

Key takeaways

Unit 4 in six lines.

1

Ethics is commercial risk management

Fines, reputation, trust and market access.

2

Privacy comes down to purpose, minimisation and consent

Collect less, use it only as promised, delete it on time.

3

Most leaks are shortcuts, not attacks

Approve safe tools so people do not go around you.

4

Bias is inherited from data and must be measured

Test by group, review with mixed teams, allow appeals.

5

Regulation follows risk, not technology

The same model is unrestricted in one use and banned in another.

6

Accountability cannot be automated

AI recommends. A named person decides and answers for it.



Thank you

Next up — Unit 5: strategy, building AI-based business solutions, and how leaders make teams dramatically more effective.

QUESTIONS TO THINK ABOUT

- What is the riskiest way AI is being used in your organisation today?
- Which of your decisions should never be fully automated?
- Who owns AI governance where you work - and do they know it?